Transcriptomic immune profiling identifies checkpoint-associated immune states in lung squamous cell carcinoma.

Agata Gabara
agatagabara@gmail.com
Independent Researcher
Koluszki, Poland

Abstract

Objectives

To characterize the immune landscape of lung squamous cell carcinoma (LUSC) and investigate the relationship between immune activation, checkpoint signaling, and exhaustion-associated immune states using transcriptomic immune profiling.

Methods

Transcriptomic data from 502 primary LUSC tumors from The Cancer Genome Atlas (TCGA-LUSC) were analysed using immune gene signatures representing tumor inflammation (TIS), cytotoxic T-cell activity, tertiary lymphoid structures (TLS), and T-cell exhaustion. Immune states were defined using median-based signature thresholds. Immune landscape organization was visualized using Uniform Manifold Approximation and Projection (UMAP) based on immune-related gene expression. Associations between immune activation, checkpoint signalling, and overall survival were evaluated using correlation and Kaplan–Meier analyses. Key findings were independently validated in the GEO GSE37745 LUSC cohort.

Results

Transcriptomic immune profiling identified heterogeneous immune states across LUSC tumors, including immune-cold, cytotoxic-high/exhaustion-low, cytotoxic-low/exhaustion-high, and cytotoxic-high/exhaustion-high phenotypes. Immune activation demonstrated a strong positive association with checkpoint signaling in the TCGA-LUSC cohort (Pearson $r = 0.88$, $p < 0.001$), which was independently reproduced in the GEO validation cohort (Pearson $r = 0.84$, $p <$

0.001). UMAP analysis revealed structured immune organization across tumors and separated immune-cold from highly inflamed tumors. Exhaustion-associated immune states consistently exhibited elevated checkpoint signaling. Survival analyses did not demonstrate statistically significant overall survival differences between immune activation groups or exhaustion states.

Conclusion

These findings define a transcriptomic framework for characterizing immune heterogeneity in LUSC and demonstrate a reproducible association between immune activation and checkpoint signaling across independent cohorts. The results support a model in which highly inflamed tumors exhibit coordinated activation of inhibitory immune regulatory programs associated with T-cell exhaustion and immune escape.

Keywords

lung squamous cell carcinoma; LUSC; tumor immune microenvironment; T-cell exhaustion; immune checkpoint signaling; transcriptomic profiling; tumor inflammation signature

List of Abbreviations

LUSC – Lung squamous cell carcinoma

TCGA – The Cancer Genome Atlas

GEO – Gene Expression Omnibus

TIS – Tumor inflammation signature

TLS – Tertiary lymphoid structures

UMAP – Uniform Manifold Approximation and Projection

OS – Overall survival

PD-1 – Programmed cell death protein 1

PD-L1 – Programmed death-ligand 1

CTLA-4 – Cytotoxic T-lymphocyte-associated protein 4

TME – Tumor microenvironment

Methods

Data acquisition and preprocessing

Transcriptomic and clinical data for lung squamous cell carcinoma (LUSC) were obtained from The Cancer Genome Atlas (TCGA-LUSC) through the UCSC Xena platform. Gene expression data were downloaded as normalized transcriptomic matrices and matched with corresponding clinical annotations. Only primary tumor samples were included in the analysis. Normal tissue samples were excluded. After filtering and patient-level matching, 502 primary LUSC tumors with available transcriptomic profiles were included in the final cohort.

An independent validation cohort was obtained from the Gene Expression Omnibus (GEO) database using dataset GSE37745. Samples annotated as squamous cell carcinoma were selected for external validation analyses. Probe-level microarray data were mapped to gene symbols using platform annotation files, and probe values mapping to the same gene were averaged.

Immune gene signatures

Immune-related transcriptomic signatures were defined based on previously reported markers of tumor inflammation, cytotoxic T-cell activity, tertiary lymphoid structures (TLS), T-cell exhaustion, and immune checkpoint signaling. Signature scores were calculated as the mean z-score-normalized expression of genes included in each signature.

The tumor inflammation signature (TIS) included the following genes:
CXCL9, CXCL10, CXCL11, IDO1, STAT1, HLA-DRA, IFNG, CD8A, GZMB, and PRF1.

The cytotoxicity signature included:
CD8A, CD8B, NKG7, PRF1, GNLY, and GZMB.

The exhaustion signature included:
PDCD1, CTLA4, LAG3, TIGIT, HAVCR2, TOX, and ENTPD1.

The TLS signature included:
CCL19, CCL21, CXCL13, CCR7, CXCR5, CD79A, and MS4A1.

The checkpoint signaling signature included:
PDCD1, CD274, CTLA4, TIGIT, LAG3, HAVCR2, IDO1, and ENTPD1.

Transcriptomic normalization

Gene expression values were z-score normalized across samples for each gene prior to signature calculation. Signature scores were calculated by averaging normalized expression values of genes present in each signature.

Immune state classification

Tumors were stratified into immune activation groups using the median TIS score across the TCGA-LUSC cohort. Samples with TIS scores greater than or equal to the cohort median were classified as "HOT," whereas samples below the median were classified as "COLD."

Immune states were further defined using median cytotoxicity and exhaustion scores. Four transcriptomic immune states were identified:

1. Cytotoxic-high / exhaustion-high
2. Cytotoxic-high / exhaustion-low
3. Cytotoxic-low / exhaustion-high
4. Cold

Dimensionality reduction and immune landscape visualization

To visualize immune landscape organization across tumors, Uniform Manifold Approximation and Projection (UMAP) was performed using expression values from immune-related genes included across all analyzed signatures. Prior to dimensionality reduction, expression values were standardized using z-score scaling. UMAP was performed using Euclidean distance with n_neighbors = 15 and min_dist = 0.3.

Survival analysis

Overall survival (OS) was calculated using days to death or days to last follow-up. Kaplan–Meier survival analyses were performed to compare survival between immune activation groups and exhaustion states. Statistical significance was assessed using the log-rank test.

Statistical analysis

Associations between immune activation and checkpoint signaling were evaluated using Pearson and Spearman correlation analyses. Statistical analyses and visualization were performed using Python (version 3.12) with pandas, NumPy, SciPy, seaborn, matplotlib, lifelines, and umap-learn packages.

Results

Transcriptomic immune profiling reveals heterogeneous immune states in TCGA-LUSC

A total of 502 primary TCGA-LUSC tumors were included in the final transcriptomic analysis after exclusion of normal tissue samples and patient-level matching. Immune-related transcriptomic signatures representing tumor inflammation (TIS), cytotoxicity, tertiary lymphoid structures (TLS), T-cell exhaustion, and checkpoint signaling were calculated across all tumors.

TCGA-LUSC cohort characteristics

Variable	Value
Total tumors	502
Patients with survival data	495
Median age	XX
Sex	XX male / XX female
Vital status	XX living / XX deceased
HOT tumors	251
COLD tumors	251
Cytotoxic-high/exhaustion-high	217
Cytotoxic-high/exhaustion-low	34
Cytotoxic-low/exhaustion-high	34
GEO validation cohort	66

Table 1. Clinical and transcriptomic characteristics of the TCGA-LUSC cohort.

Median-based stratification of TIS scores classified tumors into immune “HOT” (n = 251) and “COLD” (n = 251) groups. Additional stratification using cytotoxicity and exhaustion signatures identified four major immune states:

1. Cytotoxic-high / exhaustion-high (n = 217),
2. Cytotoxic-high / exhaustion-low (n = 34),

3. Cytotoxic-low / exhaustion-high (n = 34),
4. Cold tumors (n = 217).

These findings demonstrate substantial immune heterogeneity across LUSC tumors and support the presence of distinct exhaustion-associated immune phenotypes.

Immune-related gene expression defines structured transcriptomic immune landscapes

To visualize transcriptomic immune organization across tumors, UMAP dimensionality reduction was performed using immune-related genes derived from all analyzed signatures. The resulting immune landscape demonstrated structured organization of tumors according to immune activation and exhaustion-associated states.

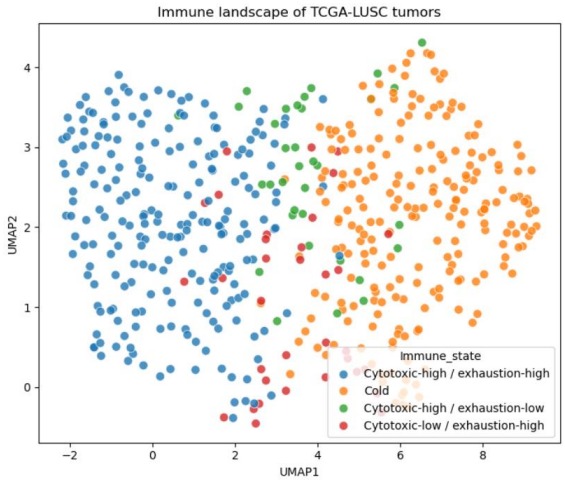

Figure 1. Transcriptomic immune landscape of TCGA-LUSC tumors.

UMAP projection generated from immune-related gene expression signatures demonstrates structured immune organization across 502 TCGA-LUSC tumors. Tumors segregate according to transcriptomic immune states, including immune-cold and exhaustion-associated inflamed phenotypes.

Immune-cold tumors formed transcriptionally distinct regions, whereas cytotoxic and exhaustion-associated tumors occupied more inflamed immune regions. Intermediate immune states were observed between highly inflamed and immune-cold tumors, consistent with continuous immune variation across the cohort.

Immune activation is strongly associated with checkpoint signaling

Correlation analyses demonstrated a strong positive association between immune activation and checkpoint signaling across TCGA-LUSC tumors. TIS scores strongly correlated with checkpoint signature activity (Pearson r = 0.88, $p < 0.001$; Spearman rho = 0.88, $p < 0.001$).

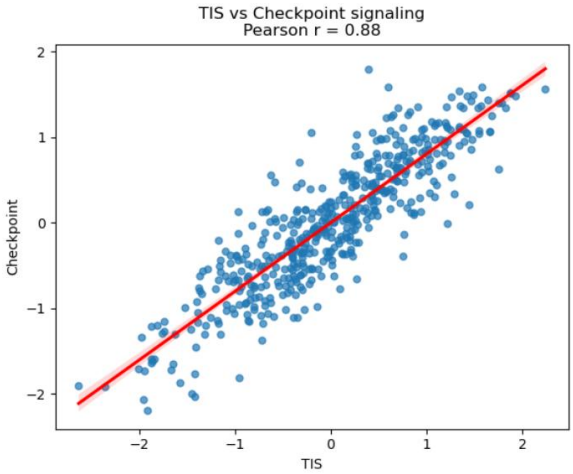

Figure 2. Association between immune activation and checkpoint signaling in TCGA-LUSC.

Scatter plot illustrating the relationship between tumor inflammation signature (TIS) and checkpoint signaling across 502 TCGA-LUSC tumors. Tumors with elevated immune activation exhibit increased checkpoint signaling activity.

Tumors with elevated cytotoxicity and exhaustion signatures consistently exhibited higher checkpoint signaling activity, supporting coordinated activation of inflammatory and inhibitory immune regulatory programs.

Independent GEO validation confirms the association between immune activation and checkpoint signaling

To validate the robustness of the observed immune associations, an independent external validation cohort was analyzed using GEO dataset GSE37745. A total of 66 squamous cell carcinoma samples were included in the validation analysis.

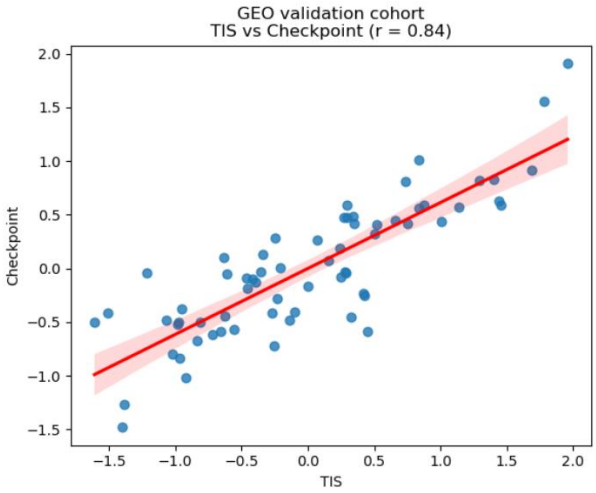

Figure 3. Independent validation of immune activation and checkpoint signaling association in the GEO GSE37745 cohort.

The strong positive association between tumor inflammation signature (TIS) and checkpoint signaling was independently reproduced in the external GEO validation cohort including 66 squamous cell carcinoma samples.

Immune signature scoring reproduced the strong association between immune activation and checkpoint signaling observed in TCGA-LUSC. TIS and checkpoint signaling remained strongly correlated in the GEO cohort (Pearson $r = 0.84$, $p < 0.001$), confirming reproducibility across independent transcriptomic datasets.

Cohort	Pearson r	Pearson p	Spearman rho	Spearman p
TCGA-LUSC	0.88	<0.001	0.88	<0.001
GEO GSE37745	0.84	<0.001	XX	<0.001

Table 2. Correlation between immune activation and checkpoint signaling across independent LUSC cohorts.

Strong positive associations between tumor inflammation signature (TIS) and checkpoint signaling were reproducibly observed in both the TCGA-LUSC discovery cohort and the independent GEO GSE37745 validation cohort.

Survival analysis reveals no significant differences between immune activation states

Kaplan–Meier analyses were performed to evaluate the association between immune activation states and overall survival. No statistically significant differences in overall survival were

observed between HOT and COLD tumors (log-rank p = 0.71) or between exhaustion-high and exhaustion-low tumors (log-rank p = 0.76).

Although immune activation and checkpoint signaling demonstrated strong transcriptomic associations, these findings suggest that immune inflammatory states alone may not fully predict survival outcomes in LUSC.

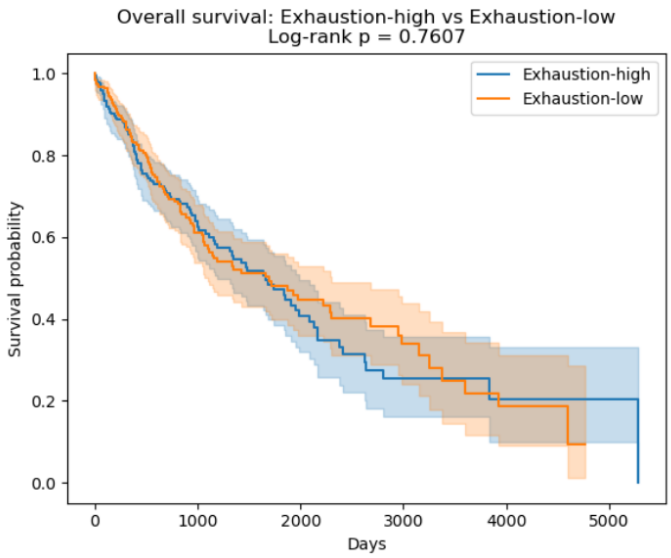

Figure 4. Kaplan–Meier survival analysis according to exhaustion-associated immune states.

Overall survival analysis comparing exhaustion-high and exhaustion-low tumors in the TCGA-LUSC cohort demonstrated no statistically significant survival differences between transcriptomic immune states.

Discussion

In this study, we performed transcriptomic immune profiling of 502 TCGA-LUSC tumors and identified heterogeneous immune states associated with inflammatory activation, cytotoxic signaling, and exhaustion-related immune programs. By integrating immune gene signatures with dimensionality reduction and validation analyses, we define a transcriptomic framework for characterizing immune organization in LUSC.

A major finding of this study is the robust association between immune activation and checkpoint signaling across independent cohorts. In both TCGA-LUSC and the external GEO validation cohort, tumors with elevated inflammatory and cytotoxic immune signatures consistently exhibited increased expression of inhibitory checkpoint molecules. These findings

support the concept that highly inflamed tumors frequently co-activate immune regulatory pathways associated with T-cell dysfunction and adaptive immune resistance.

Importantly, the association between immune activation and checkpoint signaling was highly reproducible across datasets, suggesting that this relationship represents a stable feature of the LUSC immune microenvironment rather than a cohort-specific observation. While checkpoint expression is expected to increase in activated T cells, the strong and consistent coupling observed across independent cohorts indicates coordinated immune regulatory programs associated with inflamed tumor states.

Our analysis also identified heterogeneous exhaustion-associated immune states within LUSC. Tumors classified as cytotoxic-high/exhaustion-high demonstrated the greatest checkpoint signaling activity, whereas immune-cold tumors showed minimal inflammatory and checkpoint activity. These findings are consistent with previous studies describing coexistence of inflammatory activation and inhibitory immune regulation in chronically stimulated tumor immune microenvironments.

Notably, survival analyses did not demonstrate statistically significant differences between immune activation groups or exhaustion-associated states. This observation suggests that transcriptomic inflammatory states alone may not fully determine clinical outcomes in LUSC and that additional biological, genomic, and therapeutic factors likely contribute to survival heterogeneity. Nevertheless, the observed immune programs may remain biologically relevant for understanding mechanisms of immune regulation and potential immunotherapy responsiveness.

Several limitations should be considered. First, the analyses were based on bulk transcriptomic datasets and therefore do not resolve cell-type–specific or spatial immune interactions within the tumor microenvironment. Second, the study primarily evaluates transcriptomic associations and does not establish direct causal relationships between immune activation and exhaustion-associated programs. Third, although external validation was performed using an independent GEO cohort, additional validation using single-cell transcriptomics, spatial profiling, or protein-level assays would further strengthen the biological interpretation of the identified immune states.

Although external validation was performed, prospective immunotherapy-treated cohorts were not available for direct therapy-response evaluation.

In conclusion, this study defines a reproducible transcriptomic framework for immune characterization in LUSC and demonstrates a strong association between immune activation and checkpoint signaling across independent cohorts. These findings support the presence of coordinated inflammatory and inhibitory immune programs in highly inflamed tumors and provide a basis for future studies investigating immune stratification and immunotherapy-related biomarkers in LUSC.

Declarations

Ethics approval and consent to participate

This study was based exclusively on publicly available, de-identified transcriptomic and clinical datasets obtained from The Cancer Genome Atlas (TCGA) and Gene Expression Omnibus (GEO). No new human participants or patient samples were recruited. According to institutional and TCGA/GEO data usage policies, additional ethical approval and informed consent were not required for this secondary analysis of publicly available data.

Consent for publication

Not applicable.

Author contributions

Agata Gabara conceived and designed the study, performed bioinformatic and statistical analyses, interpreted the data, generated figures, and wrote and revised the manuscript. The author approved the final version of the manuscript.

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Conflict of interest

The author declares no conflicts of interest.

Availability of data and materials

All datasets analyzed in this study are publicly available through The Cancer Genome Atlas (TCGA-LUSC) and the Gene Expression Omnibus (GEO; accession number GSE37745). Processed data, immune gene signatures, and analysis scripts are available in the project repository: https://github.com/ag48665/lusc-immune-escape-analysis

Code availability

All code used for transcriptomic processing, immune signature scoring, dimensionality reduction, statistical analysis, and visualization is available in the project repository: https://github.com/ag48665/lusc-immune-escape-analysis

AI statement

Artificial intelligence (AI)-assisted tools were used exclusively for language refinement and editorial assistance during manuscript preparation. No scientific data, analyses, results, or conclusions were generated or modified using AI systems.